

ROBOTIC WELDING · ILUO SKILL PATH

Technical Setter Interns

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Mechatronics Engineering · 9th term

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Tenneco Aguascalientes · Clean Air · September 25, 2026

TENNECO

OUR VALUES

WIN

"We must earn the trust of our employees and customers."

**Nobody hands an intern a welding torch.
You earn it.**

TENNECO CORE VALUES

RADICAL CANDOR



We must engage in productive debate and create constructive tension. Be open and brutally honest, with positive intent. Truth is harmony.

SIMPLIFY



We must reject bureaucracy, minimize layers, and eliminate silos. Be clear, break down barriers, put together a plan, and focus on execution.

ORGANIZATIONAL VELOCITY



We must build a global team and an organizational structure that facilitates fast decision making, speed of execution matters.

TENACIOUS EXECUTION



We must drive an accountability and ownership mindset that is the bedrock of our culture, and the foundation for all we do.

WIN



We must win the trust and confidence of our employees and customers.

TENNECO

Capable on the floor, not yet on paper

WHAT I LEARNED

9 of 9

Level I topics studied, none of them validated yet

WHERE IT CAME FROM

616

pages of manual and customer standard, audited page by page

WHAT I NEED

4

commitments, and none of them cost money

By January 1 I can be certified at Levels I and L.

WHERE I WORK

Few welding cells, and they run stable

PRODUCT

**Exhaust
systems**

PROCESS

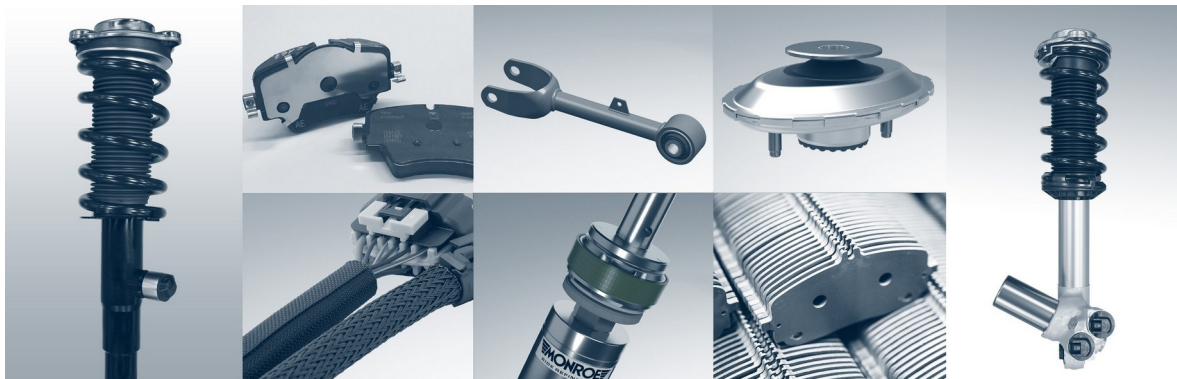
**GMAW
(MIG/MAG)**

ROBOTS

Yaskawa

EQUIPMENT

SKS, Miller



**So I could not learn by waiting for a
breakdown.**

My hours went wherever the work was.

Four levels, and half the hours are for teaching

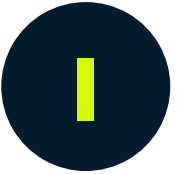


Every level has its own gate: a written exam, a validation on the floor, a practical exam, an implementation review.

LEVEL I IN DETAIL

Nine topics, one exam, one passing mark

4 hours · written exam · minimum 8.0



01

Welding safety

02

GMAW equipment

03

Metal transfer modes

04

Process variables

05

GMAW discontinuities

06

Joint geometry

07

Welding positions

08

Welding symbols

09

Welding control documents

Twenty of the forty ILUO hours sit above this level, but none of them open until this exam is passed.

Protecting myself, and the person next to me

Welding safety, per ANSI Z49.1

RADIATION

Screens, helmet and welder's clothing. Arc flash does not spare the person who is only watching.

FUMES AND GAS

The biggest factor under my own control is where I put my head relative to the fume column.

COMPRESSED GAS

Cylinders upright and secured, always. Cap on before moving one.

THE ROBOTIC CELL

Emergency stops, interlocks, teach mode at reduced speed, and lockout before anyone goes in.

The cell safety column is the one neither of my two study documents covers.

Five blocks, and the four that wear out

GMAW equipment and process variables

THE EQUIPMENT

Power source, constant voltage, DCEP

Wire feeder and drive rolls

Torch

Gas supply and flowmeter

Work lead

WHAT I CHANGE MYSELF

Contact tip

Nozzle

Diffuser

Liner

Drive rolls

There is no amperage knob. I set wire feed speed and voltage, and the current follows.

THE OTHER MANUAL PROCESS

TIG, where nothing about MIG applies

The plant runs two manual processes, and they share almost no parts

NON-CONSUMABLE ELECTRODE

Tungsten only holds the arc. Filler goes in by hand, as a rod.

100 % ARGON

Any CO₂ would consume the tungsten. 15 to 25 CFH, read on the flowmeter and nowhere else.

ITS OWN CONSUMABLES

Collet, collet body, cup and back cap. None of the MIG set fits.

A tungsten that touches the pool is cut back and reground, never cleaned.

How the metal crosses the arc

Four transfer modes, and what picks between them

SHORT CIRCUIT

Low current

The wire touches the work. Lowest heat, for thin sheet. Its risk is lack of fusion.

GLOBULAR

Mid current

Large drops fall by gravity. Heavy spatter, so we avoid it.

SPRAY

High current

Fine droplets, no contact. Needs at least 80 % argon. Flat and horizontal only.

PULSED

Controlled

The source alternates high and low. Spray quality at low heat input.

Changing the transfer mode is an essential variable. It invalidates the procedure.

Reading the print: joint, position, symbol

Three short topics that always travel together

JOINT GEOMETRY

Butt, lap, T, corner and edge.
The throat is what carries the load, not the leg.

POSITIONS

1G to 4G for groove welds, 1F to 4F for fillet welds. Cummins adds P for plug and S for slot.

SYMBOLS

AWS A2.4 by default. Arrow side goes below the reference line, the other side above.

Our prints carry two things AWS does not: the weld class and the WDR, both in the tail of the symbol.

If no class is marked, it is Class 2 by default.

A defect is a discontinuity that went too far

Discontinuities, acceptance limits and the documents that set them

WHAT WE SEE IN THE CELL

Porosity, from poor gas coverage
Undercut, from too much current or speed
Lack of fusion, from too little heat
Spatter, burn-through, cracks

WHAT SETS THE LIMIT

The weld class on the print, 1, 2 or 3
Class 1 allows no undercut at all
Cracks are never allowed, in any class
The WPS is the control plan

The same bead passes in Class 3 and fails in Class 1.

WHAT THE CUSTOMER ADDS

The customer turns that theory into numbers

From the Cummins standard for aftertreatment and exhaust parts

$\pm 5 \%$

wire feed speed, current and voltage

$\pm 10 \%$

travel speed

10 – 16 mm

contact tip to work distance,
every GMAW weld

Outside the band, the procedure has to be revalidated.

Source: CES-S-MANF-150, sections 9 and 13.

HOW I AM PREPARING

616 pages, and five gaps neither of them closes

CWI MANUAL, COMIMSA

440 pages

Covers five of the nine topics well: symbols, joint geometry, discontinuities, positions and safety.

CES-S-MANF-150, CUMMINS

176 pages

Turns that theory into numbers I can be held to: essential variables, work angle, CTWD, acceptance limits.

WHAT NEITHER ONE COVERS

Robotic cell safety

Constant voltage V-A curve

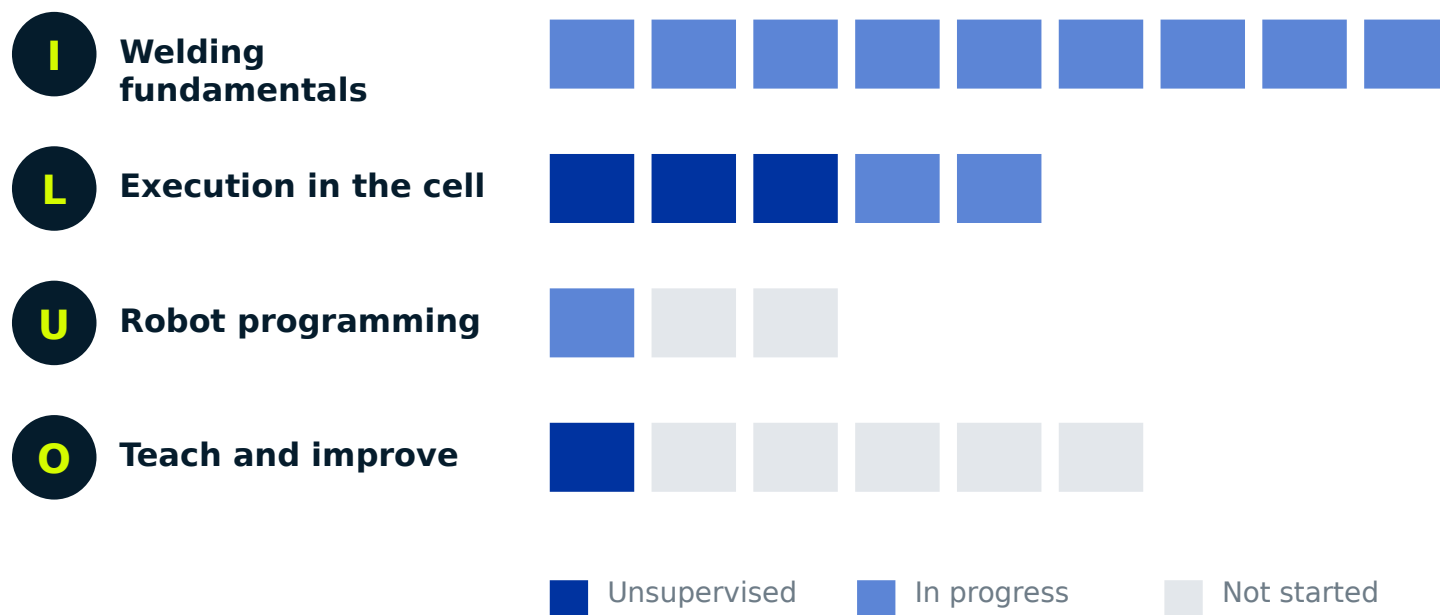
Transition current

ISO 2553 symbols

The Tenneco parameter sheet

I mapped both documents page by page. The gaps are mine to close, and I know exactly what they are.

Four skills unsupervised, none of them validated



WHAT I DO UNSUPERVISED

Consumable changes

Essential variable adjustment

Welding symbol interpretation

Closing ANDON orders

I learned Level I in the cell, not in a classroom. The exam has not happened, so nothing is validated yet.

Three levels ahead, and what each one asks



Execution in the cell

Validate parameters against the WPS and adjust the essential variables on my own.

Validation on the floor



Robot programming

Create a welding program in SKS and assign its parameters.

Practical exam



Teach and improve

Train a level 2 or 3 technician and take one improvement idea to implementation.

Implementation review

Twenty of the forty hours live in these three levels, and none of them open until Level I is signed.

60% of my hours, and none of it on the matrix

WHAT IT IS

[describe the integration in one line]

MY ROLE

[what you own day to day]

WHAT IT TAUGHT ME

[three skills outside the matrix]

STATUS

[% complete, next milestone]



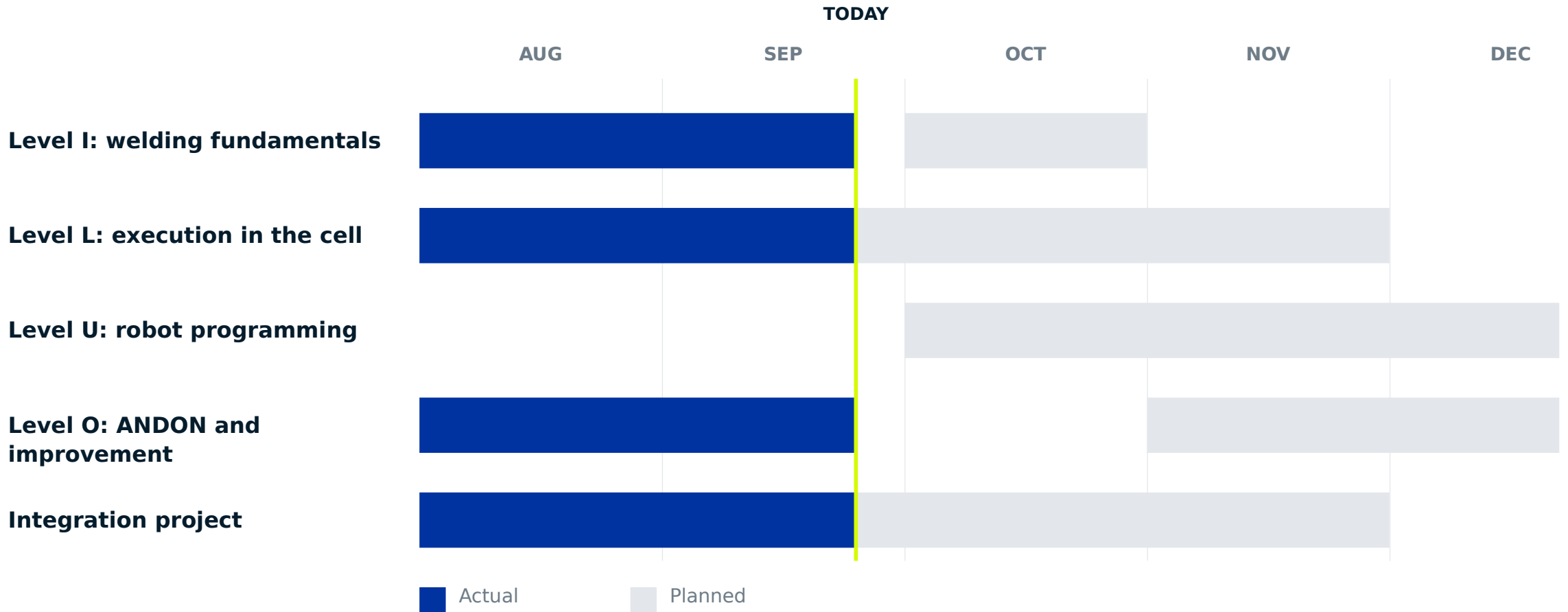
WHY IT BELONGS HERE

It is real engineering work. It simply does not appear anywhere on the ILUO matrix, so the scorecard cannot see it.

TIMELINE

14 weeks left, and Levels U and O are still ahead

August 1, 2026 → January 1, 2027



Four commitments, and none of them cost money

COMMITMENT	WHAT I NEED	WHEN
1 Sit the Level I written exam Validates Level I	A date on the calendar	October
2 Document parameter validations vs. the WPS Closes Level L	Access to the parameter sheets	Oct - Nov
3 Create one welding program in SKS, supervised Opens Level U	Cell time and supervision	November
4 Take one improvement idea to implementation First Level O evidence	A sponsor for the idea	Nov - Dec

Approve these four and I finish the program certified, not just experienced.

CONCLUSIONS

Informally capable now, certified by January

- 01** I studied all of Level I. None of it is validated yet.
- 02** The floor taught me faster than the schedule would have.
- 03** Two months ago I called the ANDON in. Now I close it.
- 04** Fourteen weeks left, and a plan for them.

Trust first, then the torch. That is how I read WIN.

Thank you

Questions?

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Diego Adair de León Márquez

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